MAT2002 Ordinary Differential Equations
First-order homogeneous ODE system with constant coefficients

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Overview

1. Fundamental matrices and matrix exponential
   - Matrix exponential
   - S-N decomposition

2. Appendix: Matrix exponential

Outline

1. Fundamental matrices and matrix exponential
   - Matrix exponential
   - S-N decomposition
2. Appendix: Matrix exponential

Fundamental matrices and matrix exponential

Recall:

$$\mathbf{x}'(t) = \mathbf{P}(t)\mathbf{x}(t) \tag{1}$$

Definition 11.1

Suppose that $\mathbf{x}^{(1)}(t), \ldots, \mathbf{x}^{(n)}(t)$ form a fundamental set of solutions for the homogeneous linear system (1). Then the matrix

$$\mathbf{\Psi}(t) = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{x}^{(1)}(t) & \mathbf{x}^{(2)}(t) & \cdots & \mathbf{x}^{(n)}(t) \\ | & | & \cdots & | \end{pmatrix} \tag{2}$$

whose columns are the vectors $\mathbf{x}^{(1)}(t), \ldots, \mathbf{x}^{(n)}(t)$ is called a **fundamental matrix** of the system (1).

Fundamental matrices and matrix exponential

Suppose that $\mathbf{x}^{(1)}(t), \ldots, \mathbf{x}^{(n)}(t)$ form a fundamental set of solutions for the homogeneous constant coefficient linear system (1), the general solution of the system (1)

$$\mathbf{x}(t) = c_1\mathbf{x}^{(1)}(t) + c_2\mathbf{x}^{(2)}(t) + \cdots + c_n\mathbf{x}^{(n)}(t)$$

can be written as

$$\mathbf{x} = \mathbf{\Psi}(t)\mathbf{c}, \tag{3}$$

where $\mathbf{c}$ is the vector with components $c_1, \cdots, c_n$. If

$$\mathbf{x}(t_0) = \mathbf{x}_0,$$

then the solution of the corresponding initial value problem is

$$\mathbf{x} = \mathbf{\Psi}(t)\mathbf{\Psi}^{-1}(t_0)\mathbf{x}_0. \tag{4}$$

Since each column of $\mathbf{\Psi}$ is a solution of the system (6), it is easy to check that $\mathbf{\Psi}$ satisfies the matrix differential equation

$$\mathbf{\Psi}' = \boldsymbol{P}(t)\mathbf{\Psi}, \tag{5}$$

where $\mathbf{\Psi}' = \frac{d\mathbf{\Psi}}{dt} = (\frac{d\psi_{ij}(t)}{dt})_{n\times n}$, $\psi_{ij}(t)$ is the (i,j)-entry of $\mathbf{\Psi}$.

Matrix exponential.

In this slide, we will use the fundamental matrix and matrix exponential to solve the following IVP

$$\boldsymbol{x}' = \boldsymbol{A}\boldsymbol{x} \tag{6}$$

$$\boldsymbol{x}(t_0) = \boldsymbol{x}_0 \tag{7}$$

Matrix exponential.

Matrix exponential. For a scalar a we have the power series expansion

$$e^{at} = 1 + \sum_{n=1}^{\infty} \frac{a^n t^n}{n!}.$$

For any $n \times n$ constant matrix $\mathbf{A}$, we can show that the following

$$\mathbf{I} + \sum_{n=1}^{\infty} \frac{\mathbf{A}^n}{n!}$$

converges to a matrix (the proof will not be shown in this course). Thus, we can defined as the matrix exponential:

$$e^{\mathbf{A}} \triangleq \mathbf{I} + \sum_{n=1}^{\infty} \frac{\mathbf{A}^n}{n!}. \tag{8}$$

And for any t, $\mathbf{I} + \sum_{n=1}^{\infty} \frac{\mathbf{A}^n \mathbf{t}^n}{n!}$ also converges, we can define

$$e^{\mathbf{At}} \triangleq \mathbf{I} + \sum_{n=1}^{\infty} \frac{\mathbf{A}^n \mathbf{t}^n}{n!}. \tag{9}$$

Matrix exponential.

Property

1. $\exp(\mathbf{O}_{n\times n}) = \boldsymbol{I}$.
2. If $\Lambda = diag(\lambda_1, \cdots, \lambda_n)$, then $e^{\Lambda} = diag(e^{\lambda_1}, \cdots, e^{\lambda_n})$.

Matrix exponential.

Theorem 11.2

1. *If* $\boldsymbol{AB} = \boldsymbol{BA}$, *then* $e^{\boldsymbol{A}+\boldsymbol{B}} = e^{\boldsymbol{A}} e^{\boldsymbol{B}}$.
2. $(e^{\boldsymbol{A}})^{-1} = e^{-\boldsymbol{A}}$
3. *For non-singular matrix* $\boldsymbol{P}$, *we have*

$$\exp(\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{-1}) = \boldsymbol{P} e^{\boldsymbol{A}} \boldsymbol{P}^{-1}.$$

Matrix exponential.

Proof.

For (1), we have (using Binomial Theorem)

$$\exp(\boldsymbol{A}+\boldsymbol{B}) = \sum_{n=0}^{\infty} \frac{(\boldsymbol{A}+\boldsymbol{B})^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{j=0}^{n} \binom{n}{j} \boldsymbol{A}^j \boldsymbol{B}^{n-j} \quad \text{By condition } \boldsymbol{AB} = \boldsymbol{BA}.$$

It follows that

$$\begin{aligned}
\exp(\boldsymbol{A}+\boldsymbol{B}) &= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{j=0}^{n} \frac{n!}{j!(n-j)!} \boldsymbol{A}^j \boldsymbol{B}^{n-j} \\
&= \sum_{n=0}^{\infty} \sum_{j=0}^{n} \frac{1}{j!(n-j)!} \boldsymbol{A}^j \boldsymbol{B}^{n-j} = \sum_{n=0}^{\infty} \sum_{j=0}^{n} \frac{\boldsymbol{A}^j}{j!} \frac{\boldsymbol{B}^{n-j}}{(n-j)!} \\
&= \sum_{j=0}^{\infty} \sum_{n=j}^{\infty} \frac{\boldsymbol{A}^j}{j!} \frac{\boldsymbol{B}^{n-j}}{(n-j)!} = \sum_{j=0}^{\infty} \frac{\boldsymbol{A}^j}{j!} \left[\sum_{n=j}^{\infty} \frac{\boldsymbol{B}^{n-j}}{(n-j)!} \right] \\
&= \sum_{j=0}^{\infty} \frac{\boldsymbol{A}^j}{j!} e^{\boldsymbol{B}} = e^{\boldsymbol{A}} e^{\boldsymbol{B}}.
\end{aligned}$$

□

Matrix exponential.

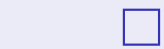
Proof.

For (2), we only need to show $e^{\boldsymbol{A}}e^{-\boldsymbol{A}} = \boldsymbol{I}$. Since

$$e^{\boldsymbol{A}}(e^{\boldsymbol{A}})^{-1} = \boldsymbol{I} = \exp(\boldsymbol{O}_{n\times n}) = e^{\boldsymbol{A}-\boldsymbol{A}} = e^{\boldsymbol{A}}e^{-\boldsymbol{A}}. \text{ since } \boldsymbol{A}(-\boldsymbol{A}) = (-\boldsymbol{A})\boldsymbol{A}.$$

For (3), we observe

$$\begin{aligned}
\exp(\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{-1}) &= \sum_{n=0}^{\infty} \frac{(\boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{-1})^n}{n!} \\
&= \sum_{n=0}^{\infty} \frac{\boldsymbol{P}\boldsymbol{A}^n\boldsymbol{P}^{-1}}{n!} \\
&= \boldsymbol{P}\left[\sum_{n=0}^{\infty} \frac{\boldsymbol{A}^n}{n!}\right]\boldsymbol{P}^{-1} \\
&= \boldsymbol{P}e^{\boldsymbol{A}}\boldsymbol{P}^{-1}
\end{aligned}$$



Matrix exponential.

Corollary

If $\mathbf{A}$ is diagonalizable, then there exists an invertible matrix $\mathbf{P}$ such that $\boldsymbol{A} = \boldsymbol{P} diag(\lambda_1, \cdots, \lambda_n)\boldsymbol{P}^{-1}$, then $e^{\mathbf{A}} = \boldsymbol{P} diag(e^{\lambda_1}, \cdots, e^{\lambda_n})\boldsymbol{P}^{-1}$.

Matrix exponential.

$$e^{\mathbf{A}t} = \sum_{p=0}^{\infty} \frac{\mathbf{A}^p t^p}{p!}.$$

Since $e^{\mathbf{A}t}$ is defined as a convergent matrix power series, it is differentiable and can be differentiated term by term (The proof will not be shown in this course). Since

$$\begin{aligned}\frac{d}{dt}(e^{\mathbf{A}t}) =& \frac{d}{dt}\sum_{p=0}^{\infty}\frac{\mathbf{A}^p t^p}{p!} = \sum_{p=0}^{\infty}\frac{d}{dt}\frac{\mathbf{A}^p t^p}{p!} = \sum_{p=1}^{\infty} p\frac{\mathbf{A}^p t^{p-1}}{p!} \\ =& \mathbf{A}\sum_{p=1}^{\infty}\frac{\mathbf{A}^{p-1}t^{p-1}}{(p-1)!} = \mathbf{A}e^{\mathbf{A}t} = \sum_{p=1}^{\infty}\frac{\mathbf{A}^{p-1}t^{p-1}}{(p-1)!}\mathbf{A} = e^{\mathbf{A}t}\mathbf{A}\end{aligned}$$

Indeed, we have the following theorem.

Theorem 11.3

$$\frac{d(e^{\mathbf{A}t})}{dt} = (e^{\mathbf{A}t})' = \mathbf{A}e^{\mathbf{A}t} = e^{\mathbf{A}t}\mathbf{A}. \tag{10}$$

Matrix exponential.

Thus, each column of $e^{\mathbf{A}t}$ is a solution for $\boldsymbol{x}' = \boldsymbol{A}\boldsymbol{x}$, moreover, $e^{\mathbf{A}t}|_{t=0} = I$. $W[e^{\mathbf{A}t}](t=0) = 1 \neq 0$. Thus, $e^{\mathbf{A}t}$ is the fundamental matrix for the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}.$$

The general solution for above ODE is

$$\mathbf{x} = e^{\mathbf{A}t}\mathbf{c}, - - - - - - - - - (*)$$

where $\mathbf{c} = [c_1, \cdots, c_n]^T$ is an arbitary constant vector.
Now we look for the solution for IVP

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \quad \mathbf{x}(t_0) = \mathbf{x}_0.$$

Substituting the initial condition for the above general solution (*), one has

$$\mathbf{x}_0 = e^{\mathbf{A}t_0}\mathbf{c}.$$

Thus, $\mathbf{c} = (e^{\mathbf{A}t_0})^{-1}\mathbf{x}_0 = e^{-\mathbf{A}t_0}\mathbf{x}_0$. Therefore,

$$\mathbf{x} = e^{\mathbf{A}t}\mathbf{c} = e^{\mathbf{A}t}e^{-\mathbf{A}t_0}\mathbf{x}_0 = e^{\mathbf{A}(t-t_0)}\mathbf{x}_0.$$

Matrix exponential for 2×2 matrix

Example 11.4

For

$$\mathbf{y}' = \mathbf{A}\mathbf{y}, \quad \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$$

with eigenvalues and corresponding eigenvectors

$$r_1 = 3, \qquad \xi_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad r_2 = -1, \qquad \xi_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix},$$

$\mathbf{A}$ is diagonalizable,

$$\mathbf{A} = \mathbf{P}\mathit{diag}(3, -1)\mathbf{P}^{-1}, \quad \mathbf{P} = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & -\frac{1}{4} \end{pmatrix}$$

Matrix exponential for 2×2 matrix

Example continue

$$e^{\mathbf{A}t} = \mathbf{P} e^{diag(3,-1)t} \mathbf{P}^{-1}$$

$$= \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} e^{3t} & 0 \\ 0 & e^{-t} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & -\frac{1}{4} \end{pmatrix} = \begin{pmatrix} \frac{e^{3t}+e^{-t}}{2} & \frac{e^{3t}-e^{-t}}{4} \\ e^{3t} - e^{-t} & \frac{e^{3t}+e^{-t}}{2} \end{pmatrix}$$

The general solution is

$$\begin{aligned} \mathbf{y}(t) &= e^{\mathbf{A}t}\mathbf{c} \\ &= \mathbf{P} e^{diag(3,-1)t} \mathbf{P}^{-1}\mathbf{c} \\ &= \mathbf{P} e^{diag(3,-1)t} \mathbf{d} \\ &= \begin{pmatrix} e^{3t} & e^{-t} \\ 2e^{3t} & -2e^{-t} \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = d_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + d_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{t} \end{aligned}$$

Fundamental matrix

Remark 1

If $\mathbf{A}$ *is diagonalizable, then* $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$*, then* $e^{\mathbf{A}t} = \mathbf{P}e^{\mathbf{\Lambda t}}\mathbf{P}^{-1}$ *is the fundamental matrix for the system*

$$\mathbf{x}' = \mathbf{A}\mathbf{x}$$

satisfying $e^{\mathbf{A}t}|_{t=0} = I$*. Moreover,* $\Psi(t) = \mathbf{P}e^{\mathbf{\Lambda t}}$ *is also a fundamental matrix.*

Matrix exponential for 3×3 matrix

Example 11.5

Find the fundamental solution matrix $\Phi(t)$ satisfying $\Phi(0) = \boldsymbol{I}$ and the general solution for system

$$\boldsymbol{x}' = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \boldsymbol{x}$$

You can verify that the three eigen-pairs of $\boldsymbol{A}$ is $(r_1, \boldsymbol{\xi}^{(1)}), (r_2, \boldsymbol{\xi}^{(2)}), (r_3, \boldsymbol{\xi}^{(3)})$, where

$$r_1 = 2, \boldsymbol{\xi}^{(1)} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}; \; r_2 = -1, \boldsymbol{\xi}^{(2)} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}; \; r_3 = -1, \boldsymbol{\xi}^{(3)} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix};$$

Hence we set

$$\boldsymbol{\Lambda} = diag(r_1, r_2, r_3), \qquad \boldsymbol{P} = \begin{bmatrix} \boldsymbol{\xi}^{(1)} & \boldsymbol{\xi}^{(2)} & \boldsymbol{\xi}^{(3)} \end{bmatrix}$$

Matrix exponential for 3×3 matrix

Example 11.6

For fundamental solution matrix satisfying $\Phi(0) = \boldsymbol{I}$, we have:

$$\Phi(t) = e^{\boldsymbol{A}t} = \boldsymbol{P}e^{\boldsymbol{\Lambda}t}\boldsymbol{P}^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} e^{-2t} & 0 & 0 \\ 0 & e^{-t} & 0 \\ 0 & 0 & e^{-t} \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} e^{2t} & e^{-t} & 0 \\ e^{2t} & 0 & e^{-t} \\ e^{2t} & -e^{-t} & -e^{-t} \end{pmatrix} \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t} & \frac{1}{3}e^{2t} - \frac{1}{3}e^{-t} & \frac{1}{3}e^{2t} - \frac{1}{3}e^{-t} \\ \frac{1}{3}e^{2t} - \frac{1}{3}e^{-t} & \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t} & \frac{1}{3}e^{2t} - \frac{1}{3}e^{-t} \\ \frac{1}{3}e^{2t} - \frac{1}{3}e^{-t} & \frac{1}{3}e^{2t} - \frac{1}{3}e^{-t} & \frac{1}{3}e^{2t} + \frac{2}{3}e^{-t} \end{pmatrix}.$$

However, if A is not diagonalizable, compute e^A is quite involving.

Semisimple-Nilpotent decomposition (S-N decomposition)

Definition 11.7

A square matrix is called Semisimple if it is diagonalizable.

And a square matrix A is called Nilpotent if there is some positive integer k s.t. $A^k = O$.

S-N decomposition (Semisimple-Nilpotent decomposition)

Theorem 11.8

(Semisimple-Nilpotent decomposition)
Let $\boldsymbol{A}$ be an $n \times n$ matrix. Then, there exist two $n \times n$ matrices $\boldsymbol{S}$ and $\boldsymbol{N}$ such that
(a) $\boldsymbol{S}$ is diagonalizable (semisimple),
(b) $\boldsymbol{N}$ is Nilpotent,
(c) $\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{N}$,
(d) $\boldsymbol{SN} = \boldsymbol{NS}$.
The two matrices $\boldsymbol{S}$ and $\boldsymbol{N}$ are uniquely determined by these four conditions.

We skip the proof for this theorem, you could check the book "*Basic Theory of Ordinary Differential Equations, Po-Fang Hsieh, Yasutaka Sibuya, Springer, 1999*" .

S-N decomposition

Although $S - N$ decomposition exists for any $n \times n$ matrix, but the construction of S and N for a general $n \times n$ matrix is related to Jordan form of the matrix (I think most students don't known). Thus, for simplicity, in this course, we only focus on the $S - N$ decomposition for 2×2 and 3×3 matrices with one single eigenvalue:

Theorem 11.9 (S-N decomposition)

Let $\boldsymbol{A}$ be a 2×2 or 3×3 matrix which has only one distinct eigenvalue r. Then $\boldsymbol{A}$ could be decomposed as $\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{N}$ such that

1. $\boldsymbol{S} = r\boldsymbol{I}$.
2. $\boldsymbol{N} = \boldsymbol{A} - \boldsymbol{S}$.
3. $\boldsymbol{N}^2 = \boldsymbol{O}$ *or* $\boldsymbol{N}^3 = \boldsymbol{O}$.
4. $\boldsymbol{SN} = \boldsymbol{NS}$

We also skip the proof for this theorem, you could check the book "*Basic Theory of Ordinary Differential Equations, Po-Fang Hsieh, Yasutaka Sibuya, Springer, 1999*" .

S-N decomposition

Fact

For $\boldsymbol{A}$ be a 2×2 or 3×3 matrix which has only one distinct eigenvalue r, we decompose it as

$$\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{N},$$

where $\boldsymbol{S} = r\boldsymbol{I}$, $\boldsymbol{N} = \boldsymbol{A} - \boldsymbol{S}$, $\boldsymbol{N}^2 = \boldsymbol{O}$ or $\boldsymbol{N}^3 = \boldsymbol{O}$. It follows that

$$e^{\boldsymbol{A}t} = e^{(\boldsymbol{S}+\boldsymbol{N})t} = e^{\boldsymbol{S}t+\boldsymbol{N}t}.$$

Since $\boldsymbol{SN} = \boldsymbol{NS}$, we derive:

$$e^{\boldsymbol{A}t} = e^{\boldsymbol{S}t} e^{\boldsymbol{N}t}$$

where

$$e^{\boldsymbol{S}t} = e^{\boldsymbol{I}(rt)} = e^{rt}\boldsymbol{I}, \quad e^{\boldsymbol{N}t} = \sum_{k=0}^{\infty} \frac{(\boldsymbol{N}t)^k}{k!} = \sum_{k=0}^{2} \frac{(\boldsymbol{N}t)^k}{k!} = \boldsymbol{I} + \boldsymbol{N}t + \frac{1}{2}\boldsymbol{N}^2t^2.$$

Hence

$$e^{\boldsymbol{A}t} = e^{rt}\left(\boldsymbol{I} + \boldsymbol{N}t + \frac{1}{2}\boldsymbol{N}^2t^2\right).$$

S-N decomposition for 2×2 matrix

Example 11.10

Find a fundamental solution matrix of

$$\boldsymbol{x}' = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \boldsymbol{x}.$$

The eigenvalues of $\boldsymbol{A}$ must satisfy:

$$\det(\boldsymbol{A} - r\boldsymbol{I}) = \begin{vmatrix} 1-r & -1 \\ 1 & 3-r \end{vmatrix} = (r-2)^2 \implies r = 2.$$

We perform the S-N decompositon for $\boldsymbol{A}$:

$$\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{N} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} + \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}$$

S-N decomposition for 2×2 matrix

Example continue

And we observe

$$e^{\boldsymbol{S}t} = e^{2t} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and

$$\boldsymbol{N}^2 = \boldsymbol{O} \implies e^{\boldsymbol{N}t} = \boldsymbol{I} + \boldsymbol{N}t = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} -t & -t \\ t & t \end{pmatrix} = \begin{pmatrix} 1-t & -t \\ t & 1+t \end{pmatrix}$$

The fundamental matrix is given by:

$$e^{\boldsymbol{A}t} = e^{\boldsymbol{S}t} e^{\boldsymbol{N}t} = e^{2t} \begin{pmatrix} 1-t & -t \\ t & 1+t \end{pmatrix}.$$

The general solution is given by:

$$\boldsymbol{x} = c_1 \begin{pmatrix} 1-t \\ t \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -t \\ 1+t \end{pmatrix} e^{2t}.$$

S-N decomposition for 3×3 matrix

Example 11.11

Find a fundamental solution matrix of

$$\boldsymbol{x}' = \begin{pmatrix} 5 & -3 & -2 \\ 8 & -5 & -4 \\ -4 & 3 & 3 \end{pmatrix} \boldsymbol{x}$$

The eigenvalues of $\boldsymbol{A}$ must satisfy:

$$\det(\boldsymbol{A} - r\boldsymbol{I}) = \begin{vmatrix} 5-r & -3 & -2 \\ 8 & -5-r & -4 \\ -4 & 3 & 3-r \end{vmatrix} = -(r-1)^3 \implies r = 1.$$

We perform the S-N decompositon for $\boldsymbol{A}$:

$$\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{N} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 4 & -3 & -2 \\ 8 & -6 & -4 \\ -4 & 3 & 2 \end{pmatrix}$$

S-N decomposition for 3×3 matrix

Example continue

And we observe

$$e^{\boldsymbol{S}t} = e^t \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and

$$\boldsymbol{N}^2 = \boldsymbol{O} \implies e^{\boldsymbol{N}t} = \boldsymbol{I} + \boldsymbol{N}t = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 4t & -3t & -2t \\ 8t & -6t & -4t \\ -4t & 3t & 2t \end{pmatrix}$$

$$= \begin{pmatrix} 4t+1 & -3t & -2t \\ 8t & -6t+1 & -4t \\ -4t & 3t & 2t+1 \end{pmatrix}$$

S-N decomposition for 3×3 matrix

Example continue

The fundamental matrix is given by:

$$e^{\boldsymbol{A}t} = e^{\boldsymbol{S}t}e^{\boldsymbol{N}t} = e^t \begin{pmatrix} 4t+1 & -3t & -2t \\ 8t & -6t+1 & -4t \\ -4t & 3t & 2t+1 \end{pmatrix}$$

Thus the fundamental solution matrix is given by:

$$\boldsymbol{x} = c_1 \begin{pmatrix} 4t+1 \\ 8t \\ -4t \end{pmatrix} e^t + c_2 \begin{pmatrix} -3t \\ -6t+1 \\ 3t \end{pmatrix} e^t + c_3 \begin{pmatrix} -2t \\ -4t \\ 2t+1 \end{pmatrix} e^t.$$

S-N decomposition for 3×3 matrix

Example 11.12

The general solution is given by

$$\boldsymbol{x}' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -3 & 2 & 4 \end{pmatrix} \boldsymbol{x}$$

The eigenvalues of $\boldsymbol{A}$ must satisfy:

$$\det(\boldsymbol{A} - r\boldsymbol{I}) = \begin{vmatrix} 1-r & 1 & 1 \\ 2 & 1-r & -1 \\ -3 & 2 & 4-r \end{vmatrix} = -(r-2)^3 \implies r = 2.$$

We perform the S-N decompositon for $\boldsymbol{A}$:

$$\boldsymbol{A} = \boldsymbol{S} + \boldsymbol{N} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} + \begin{pmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ -3 & 2 & 2 \end{pmatrix}$$

S-N decomposition for 3×3 matrix

Example continue

And we observe

$$e^{\boldsymbol{S}t} = e^{2t} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and

$$\boldsymbol{N}^2 = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 1 \\ 1 & -1 & -1 \end{pmatrix}, \boldsymbol{N}^3 = \boldsymbol{O}$$

It follows that

$$e^{\boldsymbol{N}t} = \boldsymbol{I} + \boldsymbol{N}t + \frac{1}{2}\boldsymbol{N}^2 t^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -t & t & t \\ 2t & -t & -t \\ -3t & 2t & 2t \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ -\frac{t^2}{2} & \frac{t^2}{2} & \frac{t^2}{2} \\ \frac{t^2}{2} & -\frac{t^2}{2} & -\frac{t^2}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 1-t & t & t \\ 2t - \frac{t^2}{2} & 1 - t + \frac{t^2}{2} & -t + \frac{t^2}{2} \\ -3t + \frac{t^2}{2} & 2t - \frac{t^2}{2} & 1 + 2t - \frac{t^2}{2} \end{pmatrix}.$$

S-N decomposition for 3×3 matrix

Example continue

The fundamental matrix is given by:

$$e^{\boldsymbol{A}t} = e^{\boldsymbol{S}t}e^{\boldsymbol{N}t} = e^{2t}\begin{pmatrix} 1-t & t & t \\ 2t-\frac{t^2}{2} & 1-t+\frac{t^2}{2} & -t+\frac{t^2}{2} \\ -3t+\frac{t^2}{2} & 2t-\frac{t^2}{2} & 1+2t-\frac{t^2}{2} \end{pmatrix}$$

Thus the general solution is given by:

$$\boldsymbol{x} = c_1\begin{pmatrix} 1-t \\ 2t-\frac{t^2}{2} \\ -3t+\frac{t^2}{2} \end{pmatrix}e^{2t} + c_2\begin{pmatrix} t \\ 1-t+\frac{t^2}{2} \\ 2t-\frac{t^2}{2} \end{pmatrix}e^{2t} + c_3\begin{pmatrix} t \\ -t+\frac{t^2}{2} \\ 1+2t-\frac{t^2}{2} \end{pmatrix}e^{2t}.$$

S-N decomposition for matrix

Up to now, we can solve $e^{\boldsymbol{A}t}$ when $\boldsymbol{A}$ is a 2×2 or 3×3 matrix with a single eigenvalue.

For a general $n \times n$ matrix $\boldsymbol{A}$, to compute the matrix exponential $e^{\boldsymbol{A}t}$, we need to use the Jordan form of the matrix $\boldsymbol{A}$ or find the $S - N$ composition of $\boldsymbol{A}$. However, I think most students are not familiar with Jordan form of the matrix for the general $n \times n$ matrix as it has also not been taught in the elementary linear algebra course. I put some concept related to Jordan form in the appendix, this will be taught in MAT3040. Therefore, for general $n \times n$ matrix $\boldsymbol{A}$, computing the matrix exponential $e^{\boldsymbol{A}t}$ is not required in this course. Another way to compute $e^{\boldsymbol{A}t}$ is using Putzer formula, which is also listed in the appendix and also not required.

Outline

1. Fundamental matrices and matrix exponential
   - Matrix exponential
   - S-N decomposition

2. Appendix: Matrix exponential

Appendix: Matrix exponential.

Definition 11.13 (Matrix Norm)

The *norm* of a matrix $\boldsymbol{A}$ is the number

$$||\boldsymbol{A}|| = \sup_{\boldsymbol{x} \neq \boldsymbol{0}} \frac{||\boldsymbol{A}\boldsymbol{x}||}{||\boldsymbol{x}||}$$

Or equivalently,

$$||\boldsymbol{A}|| = \sup_{||\boldsymbol{x}||=1} ||\boldsymbol{A}\boldsymbol{x}||$$

the *sphere* $||\boldsymbol{A}\boldsymbol{x}||$ for $||\boldsymbol{x}|| = 1$ is compact, and $||\boldsymbol{A}\boldsymbol{x}||$ is continuous, thus $0 < ||\boldsymbol{A}|| < \infty$.

Appendix: Matrix exponential.

Definition 11.14 (Matrix Norm)

A matrix norm on the set of all $n \times n$ matrices is a real-valued function $\|\cdot\|$, defined on this set, satisfying for all $n \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$ and all real numbers α:
(i) $\|\mathbf{A}\| \geq 0$,
(ii) $\|\mathbf{A}\| = 0$ if and only if $\mathbf{A} = \mathbf{O}$,
(iii) $\|\alpha\mathbf{A}\| = |\alpha|\|\mathbf{A}\|$,
(iv) $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$.
(v) $\|\mathbf{AB}\| \leq \|\mathbf{A}\|\|\mathbf{B}\|$.

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Definition 11.15 (Vector Norm)

Let $\mathbf{x} = [x_1, \cdots, x_n]^T \in \mathbb{R}^n$, then

$$\|\mathbf{x}\|_1 = \sum_{1 \le i \le n} |x_i|.$$

$$\|\mathbf{x}\|_\infty = \max_{1 \le i \le n} |x_i|.$$

$$\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^T\mathbf{x}} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

Theorem 11.16 (Induced Matrix Norm)

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$, if $\|\cdot\|_v$ is a vector norm defined in $\mathbb{R}^n$, then

$$\|\mathbf{A}\|_v = \max_{\mathbf{x} \in \mathbb{R}^n} \frac{\|\mathbf{A}\mathbf{x}\|_v}{\|\mathbf{x}\|_v} = \max_{\|\mathbf{y}\|_v = 1, \mathbf{y} \in \mathbb{R}^n} \|\mathbf{A}\mathbf{y}\|_v$$

defines a matrix norm. Matrix norms defined from vector norms are called the induced matrix norm.

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Theorem 11.17 (Matrix 1-Norm, 2-norm, ∞-norm)

Let $\mathbf{A} = [a_{ij}]_{n\times n} \in \mathbb{R}^{n\times n}$, then the induced matrix 1-norm, 2-norm, ∞-norm are given as follows:

$$\|\mathbf{A}\|_\infty = \max_{1\le i\le n} \sum_{1\le j\le n} |a_{ij}|.(\text{row norm})$$

$$\|\mathbf{A}\|_1 = \max_{1\le j\le n} \sum_{1\le i\le n} |a_{ij}|.(\text{column norm}).$$

$$\|\mathbf{A}\|_2 = \sqrt{\lambda_{max}(A^T A)},$$

where $\lambda_{max}(A^T A)$ is the maximum eigenvalue of $A^T A$.

The proof for this theorem is skipped. You can find the above results in the book: Richard L. Burden, J. Douglas Faires, Annete M. Burden, Numerical Analysis, 10th ed, Cengage Learning, 2015.

Appendix:Matrix exponential.

Example 11.18 (Matrix 1-Norm, 2-norm, ∞-norm)

$$A = \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix}.$$

Then

$$\|A\|_1 = \max\{1 + |-3|, |-2| + 4\} = 6$$
$$\|A\|_\infty = \max\{1 + |-2|, |-3| + 4\} = 7$$
$$\|A\|_2 = \sqrt{15 + \sqrt{221}} \approx 5.46.$$

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Definition 11.19

A sequence of $r \times r$ matrices, $\{\mathbf{A}_n\}$, is called convergent if for any given $\epsilon > 0$ there exists $N > 0$, such that

$$\|\mathbf{A}_n - \mathbf{A}_m\| < \epsilon, \qquad \forall m, n > N,$$

where the matrix norm $\|\cdot\|$ could be the 1-norm, 2-norm, or ∞-norm for matrices.

Appendix:Matrix exponential.

Theorem 11.20

Every convergent sequence of matrices $\{\mathbf{A}_n\}$ *has a limit.*

Proof.

Let a_{ij}^n, $1 \le i, j \le r$, $n = 1, 2, \cdots$, be the components of $\mathbf{A}_n$. For any given $\epsilon > 0$ there exists $N > 0$, such that

$$|a_{ij}^n - a_{ij}^m| \le \|\mathbf{A}_n - \mathbf{A}_m\| < \epsilon, \qquad \forall 1 \le i, j \le r, \quad \forall m, n > N,$$

where the matrix norm $\|\cdot\|$ could be the 1-norm, 2-norm, or ∞-norm.
Then $\{a_{ij}^n\}$ is a Cauchy sequence, and converges, *i.e.*, $a_{ij}^n \to a_{ij}$, as $n \to \infty$. Let $\mathbf{A} = [a_{ij}]$, then $\mathbf{A}$ is the limit of $\{\mathbf{A}_n\}$. □

Appendix:Matrix exponential.

Theorem 11.21

The series

$$\mathbf{I} + \sum_{n=1}^{\infty} \frac{\mathbf{A}^n t^n}{n!}. \tag{11}$$

is convergent for any finite number t, the limit matrix is defined as $e^{\mathbf{A}t}$.

Proof

Let

$$\mathbf{S}_n = \sum_{k=0}^{n} \frac{\mathbf{A}^k t^k}{k!},$$

then for $n > m$ we have

$$\|\mathbf{S}_n - \mathbf{S}_m\| = \left\| \sum_{k=0}^{n} \frac{\mathbf{A}^k t^k}{k!} - \sum_{k=0}^{m} \frac{\mathbf{A}^k t^k}{k!} \right\| = \left\| \sum_{k=m+1}^{n} \frac{\mathbf{A}^k t^k}{k!} \right\|$$

$$\leq \sum_{k=m+1}^{n} \frac{\|\mathbf{A}^k\| |t|^k}{k!} \leq \sum_{k=m+1}^{n} \frac{\|\mathbf{A}\|^k |t|^k}{k!}.$$

where the matrix norm $\|\cdot\|$ could be the 1-norm, 2-norm, or ∞-norm.

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Proof continue

Since the series

$$\sum_{k=0}^{\infty} \frac{\|\mathbf{A}\|^k |t|^k}{k!} = e^{\|\mathbf{A}\||t|}$$

converges for any $\|\mathbf{A}\||t|$ (t is finite), then for any given $\epsilon > 0$ there exists N, such that

$$\sum_{k=m+1}^{n} \frac{\|\mathbf{A}\|^k |t|^k}{k!} < \epsilon, \qquad \forall n > m > N,$$

i.e., $\{\mathbf{S}_n\}$ converges. The limit matrix is defined as $e^{\mathbf{A}t}$.

Appendix:Matrix exponential.

Theorem 2.1

$$(e^{\mathbf{A}t})' = \mathbf{A}e^{\mathbf{A}t} = e^{\mathbf{A}t}\mathbf{A}. \tag{12}$$

Clearly, for a constant matrix $\mathbf{A}$, $e^{\mathbf{A}t}$ is the fundamental matrix for the system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}.$$

Then the general solution of the above system is

$$\mathbf{x} = e^{\mathbf{A}t}\mathbf{c},$$

where $\mathbf{c}$ is the vector with components $c_1, \cdots, c_n$. Therefore, to solve the above linear system with constant coefficients, we only need to find $e^{\mathbf{A}t}$.

Appendix: Jordan form of a matrix

Note that not all matrices are diagonalizable. However, a slightly weakening of this claim is true.

Definition: Let $J_k(\lambda)$ be the $k \times k$ matrix $\begin{bmatrix} \lambda & 1 & 0 & 0 & \cdots \\ 0 & \lambda & 1 & 0 & \cdots \\ & & \cdots & & \\ 0 & 0 & 0 & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{bmatrix}$

We call each such matrix a Jordan λ-block.

A matrix J is in Jordan normal (canonical) form if

$$J = \begin{bmatrix} J_{k_1}(\lambda_1) & 0 & 0 & 0 & \cdots \\ 0 & J_{k_2}(\lambda_2) & 0 & 0 & \cdots \\ & & \cdots & & \\ 0 & 0 & 0 & J_{k_{m-1}}(\lambda_{m-1}) & 0 \\ 0 & 0 & \cdots & 0 & J_{k_m}(\lambda_m) \end{bmatrix}$$ for some integers

$k_1, \cdots, k_m$ and complex numbers $\lambda_1, \cdots, \lambda_m$.

Note that $J_1(\lambda) = (\lambda)$ and $J_k(\lambda)$ has eigenvalue λ with algebraic multiplicity k and geometric multiplicity 1.

Appendix: Jordan form of a matrix

Theorem: Any real square matrix is similar to a matrix in Jordan canonical form.

Appendix:Matrix exponential.

Theorem 2.2 (Cayley–Hamilton)

Let $\mathbf{A}$ *be an* $n \times n$ *constant matrix and let* $p(\lambda) = \det(\lambda\mathbf{I} - \mathbf{A})$. *Then* $p(\mathbf{A}) = \mathbf{0}$.

Appendix:Matrix exponential.

Proof.

Let $\mathbf{B}(\lambda)$ be the adjoint matrix of $(\lambda\mathbf{I}-\mathbf{A})$, *i.e.*, if

$$\lambda\mathbf{I}-\mathbf{A}=\begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{pmatrix},$$

then

$$\mathbf{B}(\lambda)=\begin{pmatrix} B_{11} & B_{12} & \cdots & B_{1n} \\ B_{21} & B_{22} & \cdots & B_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ B_{n1} & B_{n2} & \cdots & B_{nn} \end{pmatrix},$$

where $(\mathbf{B}(\lambda))^t$ is the matrix of cofactors of $(\lambda\mathbf{I}-\mathbf{A})$. □

Appendix:Matrix exponential.

Proof.

Then we have

$$\mathbf{B}(\lambda)(\lambda\mathbf{I}-\mathbf{A})=|\lambda\mathbf{I}-\mathbf{A}|\mathbf{I}=p(\lambda)\mathbf{I}.$$

Let

$$\mathbf{B}(\lambda)=\lambda^{n-1}\mathbf{B}_0+\lambda^{n-2}\mathbf{B}_1+\cdots+\mathbf{B}_{n-1}, \tag{13}$$

$$p(\lambda)=\lambda^n+a_1\lambda^{n-1}+\cdots+a_{n-1}\lambda+a_n, \tag{14}$$

then using (13) we get

$$\begin{aligned}\mathbf{B}(\lambda)(\lambda\mathbf{I}-\mathbf{A})=&(\lambda^{n-1}\mathbf{B}_0+\lambda^{n-2}\mathbf{B}_1+\cdots+\mathbf{B}_{n-1})(\lambda\mathbf{I}-\mathbf{A})\\ =&\lambda^n\mathbf{B}_0+\lambda^{n-1}(\mathbf{B}_1-\mathbf{B}_0\mathbf{A})+\lambda^{n-2}(\mathbf{B}_2-\mathbf{B}_1\mathbf{A})\\ &+\cdots+\lambda(\mathbf{B}_{n-1}-\mathbf{B}_{n-2}\mathbf{A})-\mathbf{B}_{n-1}\mathbf{A},\end{aligned} \tag{15}$$

□

Appendix:Matrix exponential.

Proof.

and using (14) we get

$$\begin{aligned}\mathbf{B}(\lambda)(\lambda\mathbf{I}-\mathbf{A}) =& p(\lambda)\mathbf{I}\\ =& \lambda^n\mathbf{I}+a_1\lambda^{n-1}\mathbf{I}+\cdots+a_{n-1}\lambda\mathbf{I}+a_n\mathbf{I}. \qquad (16)\end{aligned}$$

Compare the coefficients of λ^k, $k=0,1,\cdots,n$ in (15) and (16), we get $p(\mathbf{A})=\mathbf{0}$. □

Appendix:Matrix exponential.

Theorem 2.3 (Putzer)

Let $\mathbf{A}$ *be an* $n \times n$ *constant matrix with eigenvalues* $\lambda_1, \lambda_2, \cdots, \lambda_n$. *Then*

$$e^{\mathbf{A}t} = \sum_{j=0}^{n-1} r_{j+1}(t)\mathbf{P}_j, \tag{17}$$

where $\mathbf{P}_0 = \mathbf{I}$,

$$\mathbf{P}_j = (\mathbf{A} - \lambda_j \mathbf{I})(\mathbf{A} - \lambda_{j-1}\mathbf{I}) \cdots (\mathbf{A} - \lambda_1 \mathbf{I}), \qquad j = 1, 2, \cdots, n, \tag{18}$$

and $r_1(t), r_2(t), \cdots, r_n(t)$ *is the solution of the triangular system*

$$\begin{aligned} r_1' &= \lambda_1 r_1, \quad r_1(0) = 1, \\ r_j' &= r_{j-1} + \lambda_j r_j, \quad r_j(0) = 0, \qquad j = 2, \cdots, n. \end{aligned} \tag{19}$$

Appendix:Matrix exponential.

Proof.

Let $\mathbf{\Phi}(t) = \sum_{j=0}^{n-1} r_{j+1}(t)\mathbf{P}_j$. For convenience, we define $r_0(t) = 0$. Then

$$\begin{aligned}
\mathbf{\Phi}'(t) &= \sum_{j=0}^{n-1} r'_{j+1}(t)\mathbf{P}_j = \sum_{j=0}^{n-1} \{\lambda_{j+1} r_{j+1}(t) + r_j(t)\}\, \mathbf{P}_j \\
&= \sum_{j=0}^{n-1} \lambda_{j+1} r_{j+1}(t)\mathbf{P}_j + \sum_{j=0}^{n-1} r_j(t)\mathbf{P}_j \\
&= \sum_{j=0}^{n-1} \lambda_{j+1} r_{j+1}(t)\mathbf{P}_j + \sum_{j=0}^{n-2} r_{j+1}(t)\mathbf{P}_{j+1}.
\end{aligned}$$



Appendix:Matrix exponential.

Proof.

Since $\mathbf{P}_{j+1} = (\mathbf{A} - \lambda_{j+1}\mathbf{I})\mathbf{P}_j$, we have

$$\begin{aligned}\mathbf{\Phi}'(t) =& \sum_{j=0}^{n-1} \lambda_{j+1} r_{j+1}(t)\mathbf{P}_j + \sum_{j=0}^{n-2} r_{j+1}(t)(\mathbf{A} - \lambda_{j+1}\mathbf{I})\mathbf{P}_j \\ =& \lambda_n r_n(t)\mathbf{P}_{n-1} + \mathbf{A}\sum_{j=0}^{n-2} r_{j+1}(t)\mathbf{P}_j.\end{aligned}$$



Appendix:Matrix exponential.

Proof.

Suppose that the characteristic polynomial of $\mathbf{A}$ has the form

$$p(\lambda) = (\lambda - \lambda_n)(\lambda - \lambda_{n-1}) \cdots (\lambda - \lambda_1).$$

Since

$$\mathbf{P}_n = (\mathbf{A} - \lambda_n \mathbf{I})(\mathbf{A} - \lambda_{n-1}\mathbf{I}) \cdots (\mathbf{A} - \lambda_1 \mathbf{I}) = p(\mathbf{A}),$$

then $\mathbf{P}_n = 0$ by the Cayley–Hamilton Theorem. Therefore, $\lambda_n \mathbf{P}_{n-1} = \mathbf{A}\mathbf{P}_{n-1}$ and so that

$$\mathbf{\Phi}'(t) = \mathbf{A} \sum_{j=0}^{n-1} r_{j+1}(t)\mathbf{P}_j = \mathbf{\Phi}(t).$$

Moreover,

$$\mathbf{\Phi}(0) = \sum_{j=0}^{n-1} \mathbf{P}_j r_{j+1}(0) = r_1(0)\mathbf{I} = \mathbf{I}.$$

Thus, $\mathbf{\Phi}(t) = e^{\mathbf{A}t}$ by the uniqueness of the solution; □